

IV Year - I Semester

S.No	Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	PE	UGEC7T0122	Professional Elective –III a. Optical Communication and Networks (C)	3	-	-	3	30	70	100
		UGEC7T0222	b. Speech Processing (S)							
		UGEC7T0322	c. Industrial IoT(E)							
		UGEC7T0422	d. FPGA Architectures (V)							
		UGEC7T0522	e. Microwave Engineering (R)							
2	PE	UGEC7T0622	Professional Elective –IV a. Modern Satellite Communications(C)	3	-	-	3	30	70	100
		UGEC7T0722	b. Computer Vision (S)							
		UGEC7T0822	c. Industrial Automation (E)							
		UGEC7T0922	d. ASIC Design (V)							
		UGEC7T1022	e. Radar Engineering (R)							
3	PE	UGEC7T1122	Professional Elective –V a. Fundamentals of 5G (C)	3	-	-	3	30	70	100
		UGEC7T1222	b. Digital Signal Processors & Architectures (S)							
		UGEC7T1322	c. Advanced Embedded System Processors (E)							
		UGEC7T1422	d. Low Power VLSI (V)							
		UGEC7T1522	e. EMI/EMC (R)							
4	OE/JOE	UGEC7T1622	Communication Networks	2	-	2	3	30	70	100
5	OE/JOE	UGEC7T1722	Deep Learning	2	-	2	3	30	70	100
6	HSSE	UGMB7T0122	Management Science	3	-	-	3	30	70	100
7	SOC	UGEC7K1822	Amateur Radio Lab	1	-	2	2	50	-	50
8	Internship	UGEC7I1922	Industrial/ Research Internship (After Third Year)	-	-	-	3	50	-	50
			TOTAL	17	0	6	23	280	420	700
Honors/Minors (4 Credits)										

IV Year - II Semester

S.No	Category	Course Code	Course Title	L	T	P	C	IM	EM	TM
1	Major Project	UGEC8J0122	Major Project & Internship (6Months)	-	-	20	10	60	140	200
2	Seminar	UGEC8S0222	Seminar	-	2	-	2	50	-	50
			TOTAL	0	2	20	12	110	140	250
Honors/Minors (4 Credits)										

**IV YEAR
I SEMESTER**

OPTICAL COMMUNICATION AND NETWORKS

(Professional Elective –III)

Subject Code: UGEC7T0122
IV Year/ I Semester

L	T	P	C
3	0	0	3

Pre requisites: To take this course the students should have the knowledge of

- Communication Theory
- Digital Communications

Course Objectives: This course provides a full understanding of

1. The components and the design and operation of optical fiber communication systems. The principles of wavelength division multiplexed (WDM) systems.
2. The characteristics and limitations of system components like laser diodes, external modulators, optical fiber, optical amplifiers and optical receivers
3. The factors affecting the performance of both analog and digital transmission systems are studied.

SYLLABUS

UNIT-I

[12 Hrs]

OPTICAL FIBER WAVEGUIDES: Introduction, Advantages of Optical Fiber Communication, Ray Theory Transmission, Total Internal Reflection, Acceptance Angle, Numerical Aperture, Skew Rays, Step Index and Graded Index Fibers, Single Mode Fibers - Cut Off Wavelength, Mode Field Diameter and Spot Size.

UNIT – II

[10 Hrs]

SIGNAL DERADATION IN OPTICAL FIBERS: Attenuation - Absorption, Scattering and Bending losses, Core and Cladding losses. Signal Distortion in Fibers – Modal Delay, Group delay, Material Dispersion, Waveguide Dispersion, Polarization Mode Dispersion.

UNIT – III

[10 Hrs]

OPTICAL SOURCES AND PHOTO DETECTORS: LED'S - Direct and Indirect Band Gaps, LED Structures, Light Source Materials, Quantum Efficiency and LED Power. Laser Diodes - Laser Diode Modes and Threshold Conditions, Laser Diode Rate Equations, Resonant Frequencies, Narrow spectral width.

Physical Principles of Photodiodes – PIN Photodetector, Avalanche Photodiode. Structures for InGaAs APDs, Temperature Effect on Avalanche Gain.

UNIT IV**[10Hrs]**

FIBER FABRICATION AND COUPLING: Fiber Fabrication - Outside Vapor Phase Oxidation, Vapor Phase Axial Deposition, Modified Chemical Vapor Deposition, Plasma Activated Chemical Vapor Deposition. Source to Fiber Power Launching, Fiber Splicing, Optical Fiber Connectors.

Unit-V**[10Hrs]**

WDM CONCEPTS AND COMPONENTS: Overview of WDM - Operational Principles of Wavelength Division Multiplexing, WDM Standards, DWDM. Passive Optical Couplers – Two by Two Fiber Coupler. Isolators and Circulators, Fiber Grating Filters – Grating Basics, Fiber Bragg Grating, SONET.

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Blooms Level
CO 1	Explain the operation and signal attenuation of optical fibre.	II- Understanding
CO 2	Compare the operation of different Optical Sources and Optical detectors.	IV- Analyzing
CO 3	Analyze different techniques used for the fabrication of Optical Fibers and Fiber Coupling.	IV- Analyzing
CO 4	Examine WDM Concept used in Optical Communication	IV- Analyzing

Mapping of COs to POs

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	-	-

Text Books:

- T1.** Gerd Keiser, "Optical fiber Communication", McGraw Hill. 5th Edition, 2014.
- T2.** P. Chakravarthy, "Fiber Optic Communications", McGraw Hill 2015.

References:

- R1.** John Powers, "Fiber Optic Systems", Irwin Publications, 1997.
- R2.** Howes M.J., Morgen, D.V John, "Optical Fiber Communication", Wiely Publications.

SPEECH PROCESSING
(Professional Elective -III)

Subject Code: UGEC7T0222

IV Year/ I Semester

L	T	P	C
3	0	0	3

Pre-requisites: Students should have prior knowledge of

- Signals & Systems
- Random Variables & Stochastic Processes
- Digital Signal Processing

Course objectives: The objectives of this course are:

1. To provide students with the knowledge of basic characteristics of speech signal in relation to production of speech by humans.
2. To describe basic algorithms of speech analysis common to many applications of speech signal processing.
3. To give foundation for applications of speech signal processing (enhancement).

SYLLABUS

UNIT I: Speech Production

[12Hrs]

Speech processing; The speech chain: Speech production process (Lungs, Larynx and Vocal folds, Vocal tract), Speech perceiving process; Applications of Digital Speech signal processing; Phonetic representation of speech: Vowels, Diphthongs, Semi vowels, Nasals, Unvoiced fricatives, Voiced fricatives, Voiced and unvoiced stops; Digital model for speech production (Source-Filter model of speech production).

UNIT II: Time Domain Methods for Speech Processing

[10Hrs]

Time domain parameters of Speech signal; Methods for extracting the speech parameters (Energy, Average Magnitude, zero crossing Rate); Silence Discrimination using Zero crossing Rate and energy; Short Time Auto Correlation Function; Pitch period estimation using Auto Correlation Function.

UNIT III: Frequency Domain Methods for Speech Processing

[10Hrs]

Short-Time Fourier Transform (STFT); Sampling the STFT in Time and Frequency; The Speech Spectrogram;

Homomorphic speech analysis: Definition of the Cepstrum and complex Cepstrum, Short-Time Cepstrum, computation of Cepstrum, pitch extraction using homomorphic speech processing.

UNIT IV: Linear Predictive Analysis of Speech**[10Hrs]**

Linear prediction of speech and the speech model; computing the prediction coefficients: auto correlation, formulation of Linear prediction coding equations, Solution of Linear prediction coding equations; The Levinson Durbin recursion, LPC spectrum; Application of Linear prediction coding parameters: Pitch detection using Linear prediction coding parameters

UNIT V: Speech Enhancement**[10Hrs]**

Nature of Interfering Sounds; Speech Enhancement (SE) Techniques: Basic principles of Spectral Subtraction; Wiener Filtering; Wiener filtering for noise reduction; Adaptive noise cancellation, Statistical-Model-based method: Maximum-likelihood estimator for speech enhancement; Applications of speech enhancement.

Course outcomes

Upon completion of the course, students will be able to:

COs	Description	Bloom's Level
CO 1	Outline the basic characteristics of speech signal in relation to speech production and model the speech production system.	II-Understanding
CO 2	List different speech parameters.	IV-Analyzing
CO 3	Apply various algorithms for speech enhancement.	III-Appling

Mapping of COs to POs

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
CO 1	3	3												
CO 2	3	3	3											
CO 3	3	3	3											

Textbooks:

- T1.** Lawrence R. Rabiner and Ronald W. Schafer, "Introduction to Digital Speech Processing" now Publishers Inc. Hanover, USA, 2007
- T2** Douglas O Shaughnessy, "Speech Communications Human and Machine" 2nd Edition, IEEE Press, 2000.

References:

- R1** Philipos C. Loizou, "Speech Enhancement" 2nd Edition, CRC Press, Taylor & Francis Group, 2013
- R2** L. R. Rabiner and R.W. Schafer, "Digital Processing of Speech Signals" Prentice-Hall Inc. USA, 1978

INDUSTRIAL IOT

(Professional Elective-III)

Subject Code: UGEC7T0322

L	T	P	C
3	0	0	3

IV Year / I Semester

Prerequisites

- Sensors & Applications
- Modern Wireless Communications
- Embedded System with ARM

Course Objective

1. To understand the concepts of Internet of Things and its communication protocols, Applications.
2. To make students aware of resource management and security issues in Internet of Things.
3. To understand the concepts of Industrial Internet of Things, its importance and applications.

SYLLABUS

UNIT I

[12Hrs]

INTERNET OF THINGS DEFINITIONS AND FRAMEWORKS: Definition of Internet of Things, History of IoT, IoT Frameworks, Basic Nodal Capabilities, and Structural Aspects of the IoT

Communication Technologies of IOT: Sensor Technology, RFID Technology, Satellite Technology.

UNIT II

[12Hrs]

LAYER 1/2 CONNECTIVITY: WIRELESS TECHNOLOGIES FOR THE IoT:

WPAN Technologies for IoT/M2M - Zigbee/IEEE 802.15.4, Radio Frequency for Consumer Electronics (RF4CE), Bluetooth and its Low-Energy Profile, IEEE 802.15.6 WBANs, NFC, IPv6 OVER LOW-POWER WPAN (6LoWPAN).

IoT Applications / Examples: Smart Metering/Advanced Metering Infrastructure, e-Health/Body Area Networks, Home Automation and Smart Cards.

UNIT III

[10Hrs]

Designing Industrial Internet Systems: Introduction, The Concept of the IIoT, The Proximity Network, WSN Edge Node - Network Protocols, Low-Power Technologies.

Designing Low-Power Device Networks, Legacy Industrial Protocols, RS232 Serial Communications, Field Bus Technologies

UNIT IV**[10Hrs]**

Modern Communication Protocols and technologies: Industrial Ethernet, Encapsulated Field Bus, Standard Ethernet, Z-Wave, Wi-Fi Backscatter, Thread, Industrial Gateways

Middleware Transport Protocols and Software patterns: TCP/IP, UDP, Reliable Transport Protocol (RTP), CoAP (Constrained Application Protocol). MQTT, XMPP, AMQP, DDS, DTN.

UNIT V**[8 Hrs]**

Industry 4.0: Defining and the need of Industry 4.0 along with its characteristics, the value chain, Benefits to Business, Industry 4.0 Design Principles and their Building Blocks, Industry 4.0 Reference Architecture.

Smart Factories: Introduction of smart factories and its action, Importance of smart manufacturing, real-World Smart Factories, Industry 4.0: The Way Forward.

Course Outcomes: Upon completion of the course, students will be able to:

Cos	Description	Bloom's Level
CO 1	Summarize the Internet of Things and it's communication technologies.	II-Understanding
CO 2	Explain Different technologies and applications of IoT.	II-Understanding
CO 3	Examine the Designing of Industrial Internet Systems	IV-Analyzing
CO 4	Identify Modern Communication Industrial IoT.	III-Appling
CO 5	Importance of Industry 4.0 Smart factories in Industrial IoT.	V-Evaluating

Mapping of COs to POs

POs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	P 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO 1	3	3	3											
CO 2	3	3												
CO 3	3	3	3											
CO 4	3	3	3											
CO 5	3	3	3											

Text Books:

- T1.** Daniel Minoli, "Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications", Willy Publications
- T2.** "Industry 4.0: The Industrial Internet of Things", by Alasdair Gilchrist (Apress)

References:

- R1.** Parikshit N. Mahalle & Poonam N. Railkar, "Identity Management for Internet of Things", River Publishers.
- R2.** Sabina Jeschke Christian Brecher Houbing Song Danda B. Rawat Editors "Industrial Internet of Things Cybermanufacturing Systems".
- R3.** Introduction to Industry 4.0 and Industrial Internet of Things, By Prof. Sudip Misra, IIT Kharagpur.
<https://nptel.ac.in/courses/106/105/106105195/#> (or)
<https://archive.nptel.ac.in/courses/106/105/106105195/>
- R4.** <https://drive.google.com/file/d/17CPu--DdQHwUGzcbjDdNZbEcvHQ56-Cf/view>

FPGA ARCHITECTURES

(Professional Elective-III)

Subject Code: UGEC7T0422

IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisites.

- Digital Logic Design
- Digital Design through Verilog HDL

Course Objective

1. Familiarization of various complex programmable logic devices of different families.
2. to study Field programmable gate arrays and realization techniques.
3. to study different case studies using one hot design methods.

SYLLABUS

Unit-I

[10 Hrs]

Field Programmable Gate Arrays Classes: Organization of FPGAs, FPGA Programming Technologies, Programmable Logic Block Architectures, Programmable Interconnects, Programmable I/O blocks in FPGAs, Dedicated Specialized Components of FPGAs, and Applications of FPGAs

Unit-II

[10 Hrs]

Essentials and SRAM Programmable FPGA: Introduction, Programming Technology, Device Architecture, The Xilinx XC2000, XC3000 and XC4000 Architectures, FPGA implementation for combinational and sequential circuits with case studies. FPGA debugging using chip scope analyzer with case studies.

Unit-III

[10 Hrs]

Anti-Fuse Programmed FPGAs: Introduction, Programming Technology, Device Architecture.

The Actel ACT1, ACT2 and ACT3 Architectures.

UNIT-IV

[10 Hrs]

Design Applications for Systems: A Fast Video Controller, A Position Tracker for a Robot Manipulator, A Fast DMA Controller, Designing Counters with ACT Devices, Designing Adders and Accumulators with the ACT Architecture

UNIT-V

[10 Hrs]

Advanced FPGA design: Architecting Speed- High Throughput, Low Latency, Timing, Architecting Area- Rolling Up the Pipeline, Control-Based Logic Reuse,

Resource Sharing, Impact of Reset on Area, Architecting Power- Clock Control, Input Control, Reducing the Voltage Supply.

Course Outcomes: Upon completion of the course, students will be able to

CO'S	Description Blooms Level	Blooms Level
C01	Outline various architectures and FPGAs.	II-Understanding
C02	Illustrate the SRAM Programmable, Anti-Fuse Programmable FPGAs	II-Understanding
C03	Use FPGAs to implement complex digital systems.	III-Apply
C04	Optimize FPGA based digital systems design in terms of area , power and performance.	IV Analyzing

Mapping of Cos to Pos

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1					3				2					3
CO2			3								2			3
CO3			3								2			3
CO4	3	3			3									3

Text Books:

- T1.** Stephen M. Trimberger, "Field Programmable Gate Array Technology", Springer International Edition.
- T2.** Steve Kilts, "Advanced FPGA Design Architecture, Implementation, and Optimization". John Wiley & Sons, Inc., India

Reference Books

- R1.** John V. Oldfield, Richard C. Dorf, "Field Programmable Gate Arrays", Wiley India.

MICROWAVE ENGINEERING

(Professional Elective – III)

Subject Code: UGEC7T0522

IV Year / I Semester

L	T	P	C
3	0	0	3

Prerequisites: Students should have prior knowledge of

- Electromagnetic Waves.

Course Objective

1. The student should gain proficiency in using s-parameters in designing passive and active microwave circuits.
2. The student should understand the function, design, and integration of the major components oscillator, modulator, power amplifier, antenna, low-noise amplifier, filter, and mixer.

SYLLABUS

UNIT I

[12 Hrs]

RF & MICROWAVE ENGINEERING: Introduction to Microwave, Microwave

Frequencies, Microwave Applications

RECTANGULAR WAVEGUIDES: Introduction, Solutions of Wave equations in Rectangular Coordinates, Modes in Rectangular Waveguides, Impossibility of TEM Waves in Hollow Waveguides, Dominant and Degenerate Modes, Sketches of TM and TE mode fields in the cross section, Mode Characteristics - Phase and Group Velocities, Wave lengths and Impedance Relations; Power Transmission and power losses in rectangular Guide.

Unit –II

[10 Hrs]

PASSIVE MICROWAVE DEVICES: Attenuators, Microwave Hybrid Circuits: Waveguide Tees, Magic Tees (Hybrid Tees), Hybrid Rings (Rat-Race Circuits), Directional Couplers: Two-Hole Directional Couplers, Bethe-Hole Coupler, S-Matrix of a Directional Coupler, Faraday Rotation, Ferrite Components- Gyrator, Isolator, Circulator

UNIT III

[12 Hrs]

MICROWAVE LINEAR-BEAM TUBES (O TYPE): Introduction to Microwave Linear-Beam Tubes (O Type), Classification of Linear Beam Tubes (O Type), Klystrons -Velocity Modulation Process, Bunching Process, Output Power and Beam Loading (Qualitative treatment). Reflex Klystrons–Velocity Modulation, Power Output and Efficiency (Qualitative treatment).

MICROWAVE CROSSED-FIELD TUBES (M TYPE): Introduction, Classification of Cross-field tubes (M type), Magnetron Oscillators – Cylindrical Magnetron-Equations of electron motion, Cyclotron angular frequency, Power output and Efficiency.

UNIT IV**[12 Hrs]**

MICROWAVE SOLID STATE DEVICES: Introduction, Classification, Applications. Transferred Electron Devices (TEDs) – Introduction, Gunn-Effect Diodes – Background, Gunn Effect, Ridley-Watkins-Hilsum (RWH) Theory, Mode of Operation. Detector Diode and PIN Diode. Avalanche Transit Time Devices -Introduction, IMPATT Diodes-Physical Structures, Negative Resistance. TRAPATT Diodes – Physical Structures, Principle of Operation.

UNIT V**[10 Hrs]**

MICROWAVE MEASUREMENTS: Introduction, Description of Microwave Bench– Different blocks and their features, Precautions; Power Measurement–Calorimetric and Bolometer Method. Insertion loss and Attenuation measurements, VSWR Measurement, Impedance Measurement, Frequency Measurement.

Course Outcomes: Upon Completing the course, student will be able to

COs	Description	Bloom's Level
CO 1	Examine the different propagation modes of EM waves in guided structures	IV- Analyzing
CO 2	Analyze network parameters of microwave passive components.	IV- Analyzing
CO 3	Determine the output power and efficiency of various types of microwave tubes.	V- Evaluating
CO 4	Compare Transferred Electron Devices and Avalanche Transit Time devices	II- Understanding
CO 5	Measure the characteristics of microwave components	V- Evaluating

Mapping of COs to POs

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO 1	3	3												
CO 2	3	3												
CO 3	3													
CO 4	3	3	3	1									2	
CO 5	3	3											3	

Text Books

- T1.** R E Collin, "Foundation for Microwave Engineering", John Wiley & Sons, 2nd Edition, 2007
- T2.** S Y LIAO, "Microwave Devices and Circuits", PHI, 3rd Edition, 2003.

Reference Books

- R1.** Annapurna Das, Sisir K. Das, "Microwave engineering", McGraw-Hill Higher Education, 2008
- R2.** David M.Pozar, "Microwave Engineering", 3rd Edition, John Wiley & Sons, 2009.

MODERN SATELLITE COMMUNICATIONS

(Professional Elective – IV)

Subject Code: UGEC7T0622

L T P C

IV Year/ I Semester

3 0 0 3

Prerequisites: Students should have prior knowledge of

- Communication Systems - I
- Communication Systems - II

Course Objective: The objectives of this course are

- To introduce the basic concepts of satellite orbital mechanics.
- To make the students equip with knowledge on satellite segment and earth segment.
- To expose them various methods of satellite access techniques and tradeoffs that typically occurring satellite system design.

SYLLABUS

UNIT I

[10 Hrs]

ORBITAL MECHANICS AND LAUNCHERS: Orbital Mechanics, Look Angle Determination, Orbital Perturbations, Launches and Launch Vehicles, Orbital Effects in Communication System Performance.

UNIT II

[8 Hrs]

SATELLITES: Attitude and Orbit Control System (AOCS), Telemetry, Tracking, Command, and Monitoring (TTC&M), Power Systems, Communications Subsystems, Equipment Reliability and Space Qualification.

UNIT III

[8 Hrs]

SATELLITE LINK DESIGN AND MULTIPLE ACCESS: Basic Transmission Theory, System Noise Temperature and G/T Ratio, Design of Downlinks, UpLink Design, Design for Specified C/N.

Frequency Division Multiple Access (FDMA) – Intermodulation products, Time Division Multiple Access (TDMA), TDMA Frame Structure.

UNIT IV

[8 HRS]

LOW EARTH ORBIT AND NGSO SATELLITE SYSTEMS

Orbit Considerations, Coverage and Frequency Considerations, Delay and Throughput Considerations, System Considerations.

UNIT V**[8 Hrs]****SATELLITE NAVIGATION AND THE GLOBAL POSITIONING SYSTEM**

GPS Position Location Principles, GPS Receivers and Codes, GPS Navigation Message, GPS Receiver Operation, GPS C/A Code Accuracy, Differential GPS.

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Bloom's Level
CO 1	Illustrate orbital mechanics	II - Understanding
CO 2	Outline the components of satellite subsystems and satellite link design.	II- Understanding
CO 3	Identify multiple access techniques used for communicating with satellite.	III- Applying
CO 4	Illustrate the operation of GPS	II - Understanding

Mapping of COs to POs:

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO1	3	3	2											
CO2	3	2	3											
CO3	3	2												
CO4	3	2	2											

Text Books

- T1.** Timothy Pratt and Jeremy Allnutt, "Satellite Communications", 3rd Edition, Wiley Publications, 2019.
- T2.** M. Richharia, "Satellite Communications Systems", 2nd Edition, McGraw-Hill, 1999.

Reference Books

- R1.** D.C.Agarwal, "Satellite Communications", Khanna Publishers, 1989.

COMPUTER VISION

(Professional Elective -IV)

Subject Code: UGEC7T0722

IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisite

- Digital Image Processing

Course Objective

- To build detailed models of the image formation process
- To develop mathematical techniques to invert these to recover the quantities of interest

SYLLABUS

UNIT-I: Feature Detection

[10 Hrs]

Feature detection and matching Points and patches Feature Detectors Feature Descriptors Feature Matching Feature Tracking Application: Performance-driven animation Edges, Edge detection Edge linking Application: Edge editing and enhancement Lines Hough transforms, Vanishing points Application: Rectangle detection

Segmentation Active Contours Snakes, Scissors Level Sets Application: Contour tracking and rotoscoping Watershed Graph-based segmentation K-means and mixtures of Gaussians Application: Medical image segmentation.

UNIT-II: Motion Estimation

[10 Hrs]

Dense motion estimation Translational Alignment Hierarchical motion estimation Fourier-based alignment Incremental Refinement Parametric Motion Application: Video stabilization Learned motion models Spline-based motion Application: Medical image registration Optical Flow Multi-frame motion estimation Application: Video Denoising Application: De-interlacing Layered Motion Application: Frame interpolation Transparent layers and reflections

UNIT-III: Image Stitching

[10 Hrs]

Motion models Planar perspective motion Application: Whiteboard and document scanning Rotational panoramas Gap closing Application: Video summarization and compression

Cylindrical and spherical coordinates Global Alignment Bundle Adjustment Parallax Removal Recognizing Panoramas Direct vs. feature-based alignment Compositing Choosing a compositing surface Pixel selection and weighting (de-ghosting): Photomontage

UNIT-IV: Computational Photography**[10 Hrs]**

Photometric calibration Radiometric response function Noise level estimation
 Vignetting Optical blur (spatial response) estimation High dynamic range imaging
 Tone mapping Application: Flash photography Super-resolution and blur removal
 Color image demosaicing Application: Colorization Image matting and compositing
 Blue screen matting Natural image matting Optimization-based matting Smoke,
 shadow, and flash matting Video matting Texture analysis and synthesis Application:
 Hole filling and inpainting Application: Non-photorealistic rendering

UNIT-V: Recognition**[10 Hrs]**

Object detection Face detection Pedestrian detection Face recognition Eigenfaces
 Active appearance and 3D shape models: Personal photo collections Instance
 Recognition Geometric Alignment Large Databases Application: Location recognition
 Category Recognition Bag of words Part-based models Recognition with
 segmentation Application: Intelligent photo editing Context and scene understanding
 Learning and large image collections Application: Image search, Recognition
 databases and test sets

Course Outcomes: By the end of the course the student will be able to

Cos	Description	Blooms Level
CO 1	Apply algorithms on images for feature detection and motion estimation	III- Applying
CO 2	Apply algorithms to stitch images for panoramic view	III- Applying
CO 3	Apply algorithms on image and video for matting and inpainting	III- Applying
CO 4	Apply algorithms on images for various object detection and recognition tasks	III- Applying

Mapping of COs to POs

POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1									3					
CO 2						3								
CO 3						3								
CO 4							3				3			
CO 5											3			

Text Books

- T1.** Richard Szeliski, "Computer Vision: Algorithms and Applications" FIRST EDITION, Springer
- T2.** David Forsyth and Jean Ponce, "Computer Vision: A Modern Approach", Pearson; 2nd edition (14 February 2012)

Reference Books

- R1.** R.J. Schalkoff, "Digital Image Processing and Computer Vision", Wiley , 1988
- R2.** D.H.Ballard&C.M.Brown, "Computer Vision", Prentice Hall; 1st Edition (May 1,1982)

INDUSTRIAL AUTOMATION

(Professional Elective-IV)

Subject Code: UGEC7T0822

IV Year / I Semester

L	T	P	C
3	0	0	3

Prerequisites

- Fundamentals of Analog Electronics
- Digital Logic Design
- Electronic Instrumentation

Course Objective:

- To introduce Industrial Automation Concepts and its importance.
- To familiarize students with design of automated systems for Industries.

SYLLABUS

UNIT-I

[12Hrs]

Introduction to Industrial Automation: Production Systems: Facilities, Manufacturing Support System, And Automation in Production Systems: Automated Manufacturing Support Systems Computerized Manufacturing Support System, Reasons for Automating, Manual Labour in Production Systems, Automation Principles and Strategies. Introduction to Automation: - Basic Elements of Automated System Advanced Automation Functions, Levels of Automation.

UNIT-II

[10Hrs]

Hardware Components for Automation and Process Control : Characteristics and Quality Attributes of an embedded system, Application-specific embedded system-washing machine, Digital Camera & Domain -specific examples of embedded system-Automotive.

UNIT-III

[12Hrs]

COMPUTER AIDED MEASUREMENT AND CONTROL SYSTEMS: Role of computers in measurement and control, Elements of computer aided measurement and control, man-machine interface, computer aided process control hardware, process related interfaces.

Communication and networking, Industrial communication systems, Data transfer techniques, Computer aided process control software and Computer based data acquisition system, Internet of things (IoT) for plant automation.

Unit-IV**[8Hrs]**

Programmable Logic Controllers: Introduction to Micro Computers, Programmable Logic Controllers, PLC Programming, Ladder diagram, PLC Communications and Networking, PLC Selection, PLC Installation, Advantages PLCs

UNIT-V**[8 Hrs]**

DISTRIBUTED CONTROL SYSTEM: Overview of DCS, DCS software configuration, DCS communication, DCS supervisory computer tasks, DCS integration with plc and computers, features of DCS, advantages of DCS.

Course Outcomes: By the end of the course the student will be able to

COs	Description	Blooms Level
CO1	Understand different basics required for industrial automation.	II -Understanding
CO2	Understand the concepts of sensors to automate process control	II -Understanding
CO3	Apply different data transfer techniques for computer aided measurements.	III -Applying
CO4	Analyze the working PLCs	III -Analysis

Mapping of COs to POs:

POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1	3					2						3		
CO 2	3					2						3		
CO 3	3	3	2		2	2						3		
CO 4	3	3	2		2	2						3		

Text Books:

- T1.** "Automation Production Systems and Computer Integrated Manufacturing" by Mikell P. Groover, 4th Edition, Pearson Publishers.
- T2.** "Industrial Instrumentation and Control" By. S.K. Singh. The McGraw Hill Companies

Reference Books:

- R1.** Industrial control handbook by Parr, Newnes publications; 3rd edition.
- R2.** Programmable logic controller, Dunning, Delmar Thomson Learning, 2001 2nd edition.

ASIC DESIGN
(PROFESSIONAL ELECTIVE –IV)

Subject Code: UGEC7T0922
IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisites

- VLSI Design

Course Objectives

- To motivate the student to be an entry-level industrial standard ASIC designer.
- To give an understanding of issues and tools related to ASIC design

SYLLABUS

UNIT-I

[8 Hrs]

Introduction to ASICs: Types of ASICs: Full custom ASICs, Standard cell-based ASICs, Gate array based ASICs, Channeled gate array, Channel less gate array, Structured gate array, Design flow, Case study, Economics of ASICs, ASIC cell libraries.

UNIT-II

[12 Hrs]

ASIC Library Design and Programmable ASICs: Library cell design, Library architecture, Gate Array Design, Standard Cell Design, Logical effort, The Anti fuse, Static RAM, EPROM and EEPROM technology, Practical Issues, PREP benchmarks.

UNIT-III

[12Hrs]

Programmable ASIC Logic Cells: Actel ACT: ACT1, Shannon's Expansion theorem, logic module, ACT2 and ACT3 logic modules, timing model and critical path, ACTEL logic module analysis.

Xilinx LCA: XC3000 CLB, XC4000 logic block, Xilinx CLB analysis, Altera flex, Altera MAX: Logic expanders, Timing model, Power dissipation in Complex PLDs.

UNIT-IV

[10 Hrs]

Simulation, Synthesis and Static Timing Analysis: Types of Simulation: Gate level simulation, Switch level simulation, Transistor level simulation, Synthesis: Finite state machine synthesis, Memory synthesis, Performance driven Synthesis, Static Timing Analysis: Hold time, Entry delay, Exit delay and external set up time.

UNIT-V**[10 Hrs]**

Testing: The importance of test, Boundary scan test: BST cells, BST registers, Instruction decoder, TAP controller, Boundary scan controller, Faults, Fault simulation: serial fault simulation, Parallel fault simulation, Concurrent fault simulation, Automatic test pattern generation, Scan test, Built in self-test.

Course Outcomes: Upon completion of the course the students will be able to

CO'S	Description Blooms Level	Blooms Level
C01	Explain the different ASIC designs.	II -Understanding
C02	Analyze the architecture and Performance of Programmable ASICs	IV- Analyzing
C03	Elaborate the simulation, synthesis and Testing of ASICs with constraints.	V Synthesis
C04	Outline the various aspects of physical design	II -Understanding

Mapping of COs to Pos

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
C01			3		3									
C02	3	3			3								3	3
C03		3	3	3	3				3			3	3	3
C04														3

Text Books

- T1.** M.J.S.Smith, "Application Specific Integrated Circuits", Pearson Education, 2010.
- T2.** Farzad Nekoogar and Faranak Nekoogar, "From ASICs to SOCs: A Practical Approach", Prentice Hall PTR, 2003.

Reference Books

- R1.** G.Hachtel, F. Somenzi, "Logic Synthesis and Verification Algorithms" Springer; 2013
- R2.** Pak K. Chan/Samiha Mourad, "Digital Design Using Field Programmable Gate Arrays", Pearson Low Price Edition.

RADAR ENGINEERING

(Professional Elective –IV)

Subject Code: UGEC7T1022

IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisites: Students should have prior knowledge of

- Electromagnetic Waves
- Microwave Engineering

Course Objective

1. To provide an understanding of the basic concepts, operation, applications of radar systems
2. To provide an understanding of the techniques necessary to analyze the performance of radar systems.

SYLLABUS

UNIT I

[8 Hrs]

RADAR EQUATION: Radar Equation, Radar Block Diagram and Operation, Applications of Radar, Prediction of Range Performance, Minimum Detectable Signal, Receiver Noise Probability Density Functions, SNR, Integration of Radar Pulses, Radar Cross-section of Targets (simple targets-sphere, cone-sphere), Transmitter Power, PRF and Range Ambiguities, System Losses, Propagation Effects.

UNIT II

[8 Hrs]

CW AND FREQUENCY MODULATED RADAR: The Doppler Effect, CW Radar-Block Diagram, Isolation between Transmitter and Receiver, Non-Zero IF Receiver, Receiver Bandwidth Requirements, Applications of CW Radar. FM-CW Radar - Range and Doppler Measurement, Block Diagram and characteristics. FM-CW Altimeter, Multiple Frequency CW Radar.

UNIT III

[10 Hrs]

MTI AND PULSE DOPPLER RADAR: Introduction, MTI Radar with power amplifier transmitter, MTI Radar with power oscillator transmitter.

Delay line Cancellers, Multiple or staggered PRF, Range gated Doppler filters, Limitation to MTI performance, Non coherent MTI, Pulse Doppler Radar, MTI from a Moving Platform.

UNIT IV**[8 Hrs]**

TRACKING RADAR: Tracking with Radar, Sequential Lobing, Conical Scan, Monopulse Tracking Radar-amplitude comparison monopulse, Phase comparison monopulse. Tracking in range, Acquisitions.

UNIT V**[8 Hrs]**

RADAR Receivers: Lens Antennas, Phased array Antennas, Radar Receivers- Noise Figure and Noise Temperature, Mixers, Low Noise front-ends, Radar , Duplexer and Receiver Protectors.

Course outcomes

Upon completion of the course, students will be able to:

COs	Description	Bloom's Level
CO 1	Outline the concept of Radar Equation	II - Understanding
CO 2	Analyze the operation of various types of Radars	IV - Analyze
CO 3	Analyze the operation of various types of Tracking Radars	IV - Analyze
CO 4	Identify the functions of Radar Receivers.	III - Applying

Mapping of COs to POs:

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO 1	3	3												
CO 2	3	3												
CO 3	3	3												
CO 4	3													

Text Book

- T1.** Merrill I skolnik, "Introduction to Radar Systems", McGraw Hill, 2nd Edition,2007.
- T2.** G S N Raju, "Radar Engineering and Fundamentals of Navigational Aids", IK international Publishers, 2008

Reference Books

- R1.** Peyton Z Peebles Jr. (2004), "Radar Principles", John Wiley Inc.

FUNDAMENTALS OF 5G COMMUNICATIONS

(Professional Elective –V)

Subject Code: UGEC7T1122

L	T	P	C	
	3	0	0	3

IV Year/ I Semester

Pre requisites

- Modern Wireless Communications

Course Objectives: In this course it is aimed to introduce to the students about

- a comprehensive overview of the current state of the 5G landscape, likely use cases
- a wide range of technology options and potential 5G system architectures, to spectrum issues

UNIT-I

Concepts of 5G Systems: Historical Trend for Wireless Communication -Mobile Communications Generations: 1G to 4G –Evolution of LTE Technology to Beyond 4G – Pillars of 5G –Standardization Activities -Use cases and Requirements–System Concept –Spectrum and Regulations: Spectrum for 4G –Spectrum Challenges in 5G – Spectrum Landscape and Requirements –Spectrum Access Modes and Sharing Scenarios

UNIT – II

5G ARCHITECTURE :5G Architecture: Software Defined Networking –Network Function Virtualization –Basics about RAN Architecture –High-Level Requirements for 5G Architecture –Functional Architecture and 5G Flexibility –Physical Architecture and 5G Deployment

UNIT – III

Millimeter Wave Communication: Spectrum and Regulation, Channel Propagation –Hardware Technologies for mmW Systems –Deployment Scenarios.

Architecture and Mobility –Beamforming –Physical layer Techniques, Transmission Schemes

UNIT – IV

MACHINE TYPE COMMUNICATION: Use cases and Categorization of MTC, MTC Requirements, Data and Control for short packets, non-orthogonal access Protocols,

Massive MTC: Design principles, technology components, summary of mMTC, Ultra-reliable Low-latency MTC: Design principles, technology components.

UNIT – V

Device-to-Device COMMUNICATION: D2D: from 4G to 5G –Radio Resource Management for Mobile Broadband D2D–multi-hop D2D Communications for Proximity and Emergency Services –Multi-operator D2D Communication

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Bloom's Level
CO 1	Describe and explain the evolution of 5G, system concepts and spectrum challenges	II - Understanding
CO 2	Illustrate and explain the 5G functional and physical architecture and its requirements	III - Applying
CO 3	Explain the architecture, Beamforming and hardware technologies for mmW communications	II - Understanding
CO 4	Describe and explain the requirements and fundamental techniques for MTC and D2D Communication	IV - Analyzing

Mapping of COs to POs:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	3	-	-	-	-	-	-	-	-	-	-	-
CO4	3	2	2	-	-	-	-	-	-	-	-	-	-	-

Text Books

- T1.** Asif Oseiran, Jose F. Monserrat and Patrick Marsch, "5G Mobile and Wireless Communications Technology", Cambridge University Press, 2016.
- T2.** Jonathan Rodriguez, "Fundamentals of 5G Mobile Networks", Wiley, 2015

References

- R1.** Patrick Marsch, Omer Bulakci, Olav Queseth and Mauro Boldi, "5G System Design –Architectural and Functional Considerations and Long Term Research", Wiley, 2018

DIGITAL SIGNAL PROCESSORS AND ARCHITECTURE

(Professional Elective -V)

Subject Code: UGEC7T1222
IV Year/ I Semester

L	T	P	C
3	0	0	3

Pre-Requisites

- Digital Signal Processing

Course Objectives

1. To outline the architectures of different types of DSP Processors.
2. To implement basic DSP algorithms on different DSP processor.

SYLLABUS

UNIT-I

[10 Hrs.]

INTRODUCTION TO PROGRAMMABLE DSPs: Introduction, Basic Architectural Features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation Unit, Programmability and Program Execution, Speed Issues, Features of External Interfacing.

UNIT-II

[10 Hrs.]

PROGRAMMABLE DIGITAL SIGNAL PROCESSOR (TMS320C54XX): Introduction, Bus Structure, CPU and Internal memory and Memory Mapped Registers, Data Addressing Modes and Memory Space of TMS320C54XX, Program Control, Instructions and Programming, On chip Peripherals, and Interrupts of TMS320C54XX, Pipeline Operation of TMS320C54XX.

UNIT-III

[10 Hrs]

DEVELOPMENT TOOLS FOR DSP IMPLEMENTATIONS: Introduction, DSP Development Tools, DSP System Design Kit, Software for Development.

The Assembler and Assembly Source File, the Linker and Memory Allocation, the C/C++ Compiler, Code Composer Studio, DSP Software Development Example.

UNIT-IV

[10 Hrs]

ADVANCED PROCESSORS: Features of TMS320C62X Processors, Internal Architecture, CPU and Data Paths of TMS320C62X Processors, Addressing Modes, memory architecture, External memory Access, Pipeline Operation, Peripherals, **Overview of Motorola DSP563XX Processor:** - Data ALU, Multiplier Accumulator,

Address Generation Unit, Program Control Unit, JTAG TAP and On CE module, On Chip, Peripherals, and Memory internal Buses and DMA.

UNIT-V

[10 Hrs]

APPLICATIONS OF PROGRAMABLE DSP DEVICES: Introduction, DSP Based Biotelemetry system, Speech Processing System, Image processing System, DSP Based Power meter.

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Bloom's Level
CO 1	Outline the fundamentals of programmable DSPs	II-Understanding
CO 2	Explain the architectures TMS320C (54XX, 62X) and MotorolaDSP563XX Processor	II-Understanding
CO 3	Make use of IDE & DSP processors for implementation of signal processing algorithms	III-Applying
CO 4	Apply the DSP Processors in Different fields of Engineering.	IV-Analyzing

Mapping of COs to POs

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
CO 1	3													
CO 2	3		3	3										
CO 3	3		3	3	3									
CO 4	3		3	3	3	3	3							

Text Books

- T1.** Avtar Singh, S.Srinivasan "DSP Implementation using DSP microprocessor with Examples from TMS32C54XX", THAMSON 2004.
- T2.** B.VenkataRamani and M. Bhaskar, "Digital Signal Processors, Architecture, and Programming", TMH, 2004.

Reference Books

- R1.** Lapsley et al. "DSP Processor Fundamentals, Architectures & Features", S. Chand & Co, 2000.
- R2.** Jonathen Stein, "Digital signal processing", John Wiley 2005.

Advanced Embedded Systems Processors

(Professional Elective -V)

Subject Code: UGEC7T1322

IV Year / I Semester

L	T	P	C
3	0	0	3

Prerequisites

- Digital Logic Design
- Embedded Systems with ARM

Course Objectives

- To introduce the concepts of Embedded and Real-time operating systems
- To understand the design aspects of Embedded systems related to hardware, software and testing.
- To familiarize Design of Embedded System in different Domain-Specific IoT.

SYLLABUS

UNIT-I

[8 Hrs]

INTRODUCTION: Embedded systems overview, Design challenges, Embedded Processor Technology, IC Technology, Design Technology, Trade-offs.

UNIT-II

[12Hrs]

Embedded Hardware & Firmware Design and Development: Hardware Design: Analog & Digital Electronic Components, Serial Communication Devices (I2C, SPI, CAN), Electronic Design Automation (EDA) Tools, The PCB Layout Design
Firmware Design: Embedded Firmware Design approaches, Development Languages, ISR Concept, Interrupt Service Mechanism, Basic concepts Embedded C and Sample programs.

UNIT-III

[12Hrs]

Real-Time Operating System: Operating System Basics, Types of O.S, Kernel Architecture, Tasks, process and Threads, Task Scheduling.
Threads, Process Scheduling, Task Communication & Synchronization, Examples of handheld & Real-time Operating systems.

UNIT- IV

[10 Hrs]

The Embedded Product Development Life Cycle (EDLC): What is EDLC, Why EDLC, Objectives of EDLC, Different Phases of EDLC, EDLC Approaches (Modeling the EDLC).

UNIT-V**[10 Hrs]**

Developing Internet of Things: IoT Design Methodology, Domain-Specific IoTs, Case Studies - Case Study-1 (Home Automation): Smart Lighting, Smart Appliances, Intrusion Detection, Smoke/gas Detectors. Case Study-2 (Cities): Smart Parking, Smart Lighting, Smart Roads, Surveillance. Case Study-3 (Environment): Weather Monitoring, Air Pollution Monitoring, Forest Fire Monitoring, River Floods Detection. Case Study-4 (Agriculture & Lifestyle): Smart Irrigation, Green House Control, Wearable Electronics.

Course Outcomes: By the end of the course, the student will be able to

COs	Description	Blooms Level
CO1	Outline concepts of embedded systems and various Processor and IC Technologies associated with them.	II - Understanding
CO2	Build a basic model of Embedded System by studying all Hardware Components and Software Requirements required	III-Appling
CO3	Outline basic concepts of RTOS.	II – Understanding
CO4	Summarize Life Cycle of an Embedded Product Development and various Embedded System applications using IoT	II – Understanding

Mapping of COs to POs:

CO\PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	3		3									2		
CO2	2	3	3		3							3	3	3
CO3		3	3									3		
CO4		3	3		3				2			3	3	

Text Books

- T1.** Frank Vahid, Tony D. Givargis, "Embedded system Design: A Unified Hardware/Software Introduction", John Wily & Sons Inc., 2002. (Unit-I)
- T2.** Shibu. K.V, "Introduction to Embedded systems" by -Tata McGraw Hill Education Private limited, 2009. (Unit-II,III& IV)
- T3.** Vijay Madiseti and ArshdeepBahga, "Internet of Things (A Hands-onApproach)", 1st Edition, VPT, 2014. (Unit-V & VI)

References

- R1.** Embedded systems Architecture by Tammy Noergaard, Elsevier Publications, 2005.
- R2.** Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Aves and, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1 st Edition, Academic Press, 2014.

LOW POWER VLSI
(Professional Elective –V)

Subject Code: UGEC7T1422

IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisites

- VLSI Design
- Microprocessors

Course Objective:

- To expose the students to the low voltage device modeling.
- To design Low voltage, low power VLSI CMOS circuit design.

Syllabus

UNIT-I

[8 Hrs]

Needs for Low Power VLSI and Sources of power dissipation: Needs for Low Power VLSI and Sources of power dissipation: Charging and Discharging Capacitance, CMOS Leakage Current, Static Current, Principles of Low Power Design, Low Power Figure of Merits.

UNIT-II

[10 Hrs]

Low Power Design Approaches: Low-Power Design through Voltage Scaling – VTCMOS circuits, MTCMOS circuits, Architectural Level Approach – Pipelining and Parallel Processing Approaches. Switched Capacitance Minimization Approaches System Level Measures, Circuit Level Measures, Mask level Measures. Adiabatic Logic Circuits: Adiabatic Charging, Adiabatic Amplification, Adiabatic Logic Gates.

UNIT-III

[10 Hrs]

Low-Voltage Low-Power Adders: Introduction, Standard Adder Cells, CMOS Adder's Architectures – Ripple Carry Adders, Carry Look-Ahead Adders, Carry Select Adders, Carry Save Adders.

Low-Voltage Low- Power Design Techniques –Trends of Technology and Power Supply Voltage, Low-Voltage Low-Power Logic Styles.

UNIT-IV

[10 Hrs]

Low-Voltage Low-Power Multipliers: Introduction, Overview of Multiplication, Types of Multiplier Architectures, Braun Multiplier, Baugh-Wooley Multiplier, Booth Multiplier, Introduction to Wallace Tree Multiplier.

UNIT-V

[12Hrs]

Low-Voltage Low-Power Memories Basics of ROM, Low-Power ROM Technology, Future Trend and Development of ROMs, Basics of SRAM, Memory Cell, Precharge and Equalization Circuit, Low-Power SRAM Technologies, Basics of DRAM, Self-Refresh Circuit, Future Trend and Development of DRAM.

Course Outcomes: On successful completion of the course the students will be able to

CO'S	Description Blooms Level	Blooms Level
CO1	Demonstrate in depth knowledge in the limitations of low power design and different sources of power dissipation.	II -Understanding
CO2	Acquire the knowledge in various low power design approaches and techniques.	III -Applying
CO3	Analyze and design various types of low power adders.	IV-Analyzing
CO4	Apply the different low power techniques for designing multipliers and Low-Voltage Low-Power Memories.	III- Applying

Mapping of COs to POs:

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3		3									2		3
CO2	3	3										2		3
CO3			3									2		3
CO4					3							2		3

Text Books:

- T1.** Kiat-Seng Yeo, Kaushik Roy, "Low-Voltage, Low-Power VLSI Subsystems", TMH Professional Engineering
- T2.** Gary K. Yeap, "Practical Low Power Digital VLSI Design", Kluwer Academic Press, 2002.
- T3.** Sung-Mo Kang, Yusuf Leblebici, "CMOS Digital Integrated Circuits: Analysis and Design", TMH, 2011.

Reference Books

- R1.** AnanthaChandrakasan, "Low Power CMOS Design", IEEE Press/Wiley International, 1998.
- R2.** Pal, Ajit, "Low-Power VLSI Circuits and Systems", Springer, 2015.
- R3.** Kaushik Roy, Sharat C. Prasad, "Low Power CMOS VLSI Circuit Design", John Wiley & Sons, 2000
- R4.** A. Bellamour, M. I. Elamasri, "Low Power CMOS VLSI Circuit Design", Kluwer Academic Press, 1995.

EMI/EMC
(Professional Elective –V)

Subject Code: UGEC7T1522

IV Year/ I Semester

L	T	P	C
3	0	0	3

Prerequisites:

- Electromagnetic Waves
- Microwave Engineering

SYLLABUS

UNIT – I

[8 Hrs]

Natural and Nuclear Sources of EMI / EMC: History, Concepts and Definitions of EMI/EMC, Electromagnetic environment, Practical experiences and concerns, frequency spectrum conservations, An overview of EMI / EMC, Natural and Nuclear sources of EMI and Biological effects of EMI

UNIT – II

[8 Hrs]

EMI from Apparatus, Circuits and Open Area Test Sites: Electromagnetic emissions, Noise from relays and switches, non-linearity in circuits, passive intermodulation, Cross talk in transmission lines, Transients in power supply lines, Open area test sites and measurements.

UNIT – III

[10 Hrs]

Radiated and Conducted Interference Measurements and ESD: Anechoic chamber, TEM cell, GTEM Cell. Characterization of conduction currents / voltages

Conducted EM noise on power lines, Conducted EMI from equipment.

UNIT – IV

[8 Hrs]

ESD, Grounding, Shielding, Bonding, and EMI filters: ESD, Electrical fast transients / bursts, Electrical surges, Principles and types of grounding, Shielding, and bonding, Characterization of filters, Power lines filter design.

UNIT – V

[8 Hrs]

EMC Standards: Standards for EMI and EMC, MIL Standards, IEEE/ANSI standards, CISPR/IEC standards, FCC regulations, Euro norms, British Standards, EMI/EMC standards in JAPAN.

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Bloom's Level
CO 1	Understand EMC regulation and methods of eliminating interferences	II-Understanding
CO 2	Illustrate the concept of Interference Measurements	II-Understanding
CO 3	Understand the concept of filtering and shielding	II-Understanding
CO 4	Understand about electrostatic discharge and standards	II-Understanding

Mapping of COs to POs

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO 1	2	2		2					2				2	
CO 2	3	3				2	2			3	1		2	
CO 3	2								1					
CO 4	3	3		3					3					

TEXT BOOKS:

- T1.** C.R. Pal., "Introduction to Electromagnetic Compatibility", Ny John Wiley, 1992.
- T2.** "Electromagnetic Interference and Compatibility", IMPACT series, IIT-Delhi, Modules1-9.

REFERENCE BOOKS:

- R1.** Dr. V.P. Kodali, IEEE Publication, "Engineering Electromagnetic Compatibility", Printed in India by S. Chand & Co. Ltd., New Delhi, 2000.
- R2.** Henry W. Ott, "Electromagnetic Compatibility Engineering", John Wiley & Sons, 2009

Communication Networks

(Job Oriented Elective)

Subject Code: UGEC7T1622

L T P C

IV Year/ I Semester

2 0 2 3

Prerequisites: Students should have prior knowledge of

- Communication Theory
- Digital Communications

Course Objective: The objectives of this course are

- To introduce the basic concepts of Computer Networks.
- To understand the working of different layers of the Computer Networks.

SYLLABUS

Unit-I

Introduction : Uses of Computer Networks, Reference Models – OSI reference model and TCP/IP reference model.

Introduction to MATLAB Programming – import and export operations, elements, plotting, Java Classes, The Guide.

Unit-II

Physical Layer: Guided Transmission – Twisted Pairs, Coaxial Cable & Fiber Optics; Wireless Transmission – Microwave transmission.
Digital Communication System Simulation using MATLAB.

Unit-III

Data Link Layer: Data Link Layer design issues; Sliding Window protocols; Multiple Access protocols – carrier sense multiple access protocols.

Introduction to Network Topologies and Network Topologies like star, ring, mesh, simulation using MATLAB.

Unit-IV

The Network Layer: Network Layer design issues; Routing Algorithms – Shortest Path algorithm, Flooding, Distance Vector routing, Link State Routing, Multicast routing; AODV Routing.
Simulation of Network Routing algorithms using MATLAB

Unit-V

The Transport Layer: The transport service, Elements of transport protocols, the internet transport protocols: Introduction to UDP, Introduction to TCP.

The Application Layer: Domain Name System, Electronic Mail, The World Wide Web, Streaming audio and video, Voice-over-IP.

Course Outcomes: Upon completion of the course, students will be able to

COs	Description	Bloom's Level
CO 1	Illustrate the basic concept of Computer Networks	II - Understanding
CO 2	Outline the components of physical layer.	II- Understanding
CO 3	Identify various protocols used in data link layer.	III- Applying
CO 4	Discuss the operation of network layer.	II - Understanding
CO 5	Illustrate the functioning of application layer.	II- Understanding

Mapping of COs to POs:

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO1	3	3	2											
CO2	3	2	3											
CO3	3	2												
CO4	3	2	2											
CO5	3	3												

Textbooks

- T1.** Computer Networks---- Andrew S TANENBAUM, 4th Edition. Pearson Education/PHI.
- T2.** Network Modeling, Simulation and Analysis in MATLAB --- Dac-Nhuong Le, Abhishek Kumar Pandey et al, WILEY publishing, 2019.

Reference Books:

- R1.** Data Communications and Networking----- Behrouz A. Forouzan. Third Edition TMH.
- R2.** An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education.

DEEP LEARNING

(Job Oriented Elective)

Subject Code: UGEC7T1722
IV Year/ I Semester

L	T	P	C
2	0	2	3

Prerequisites

- Mathematics
- Machine Learning

Course Objectives: The objective of the course is

1. To provide exposure on the advances in the field of deep learning
2. To apply for real world problems.

SYLLABUS

UNIT I

[10 Hrs]

Introduction: Various paradigms of learning problems, Perspectives and Issues in deep learning framework, review of fundamental learning techniques. Feed forward neural network: Artificial Neural Network, activation function, multi-layer neural network.

UNIT II

[10 Hrs]

Training Neural Network: Risk minimization, loss function, back propagation, regularization, model selection, and optimization.

UNIT III

[10 Hrs]

Deep Learning: Deep Feed Forward network, regularizations, training deep models, dropouts.

Convolution Neural Network, Recurrent Neural Network, Deep Belief Network.

UNIT IV

[10 Hrs]

Probabilistic Neural Network: Hopfield Net, Boltzmann machine, RBMs, Sigmoid net, Auto encoders.

UNIT V

[10 Hrs]

Deep Learning research: Object recognition, sparse coding, computer vision, natural language processing. Deep Learning Tools: Eras, Pytorch.

Course Outcomes: Upon completion of the course the students will able to

COs	Description	Blooms Level
CO 1	Demonstrate the fundamental principles, theory and approaches of artificial neural networks.	II -Understanding
CO 2	Illustrate the various learning algorithms in artificial neural network	II -Understanding
CO 3	Illustrate the deep learning concepts	II -Understanding
CO 4	Demonstrate the probabilistic neural network concepts	II -Understanding
CO 5	Identify new application requirements in the field of computer vision.	III- Applying

Mapping of COs with POs:

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
CO 1	3													
CO 2														
CO 3				3	3						3			
CO 4												3		
CO 5	3													

Text Books

- T1.** Goodfellow, I., Bengio,Y., and Courville, A., "Deep Learning", MIT Press, 2016..
- T2.** Bishop, C. ,M., "Pattern Recognition and Machine Learning", Springer, 2006.

Reference Books

- R1.** Yegnanarayana, B., "Artificial Neural Networks", PHI Learning Pvt. Ltd, 2009.
- R2.** Golub, G.,H., and Van Loan,C.,F., Matrix Computations, JHU Press,2013.
- R3.** Satish Kumar, Neural Networks: A Classroom Approach, Tata McGraw-Hill Education, 2004.
- R4.** Ravindran, K. M. Ragsdell , and G. V. Reklaitis , "ENGINEERING OPTIMIZATION: Methods and Applications" , John Wiley & Sons, Inc., 2016
- R5.** Antoniou, W. S. Lu, "PRACTICAL OPTIMIZATION Algorithms and Engineering Applications", Springer, 2007.

MANAGEMENT SCIENCE

(Common to all branches)

Subject Code: UGMB7T0122

IV Year / I Semester

L	T	P	C
3	0	0	3

Prerequisites:

- General awareness about Principles of Management.
- To have an insight about Production and Operations Management.
- To be able to acquire knowledge about Human Resource Management, Marketing, Strategic Management.

Course Objectives:

1. To create awareness about different Managerial concepts like Management, Production, Marketing, Human Resource and Strategic Management.
2. To make the students equip with knowledge on techniques of PERT and CPM in project management.

SYLLABUS

UNIT-I

[8 Hrs]

Introduction to Management: Concept and importance of Management, Functions of management, Evaluation of Management thought, Fayol's principles of Management, Maslow's need hierarchy & Herzberg's two factor theory of Motivation, Decision making process, Designing organizational structure, Principles of Organization, Types of organization structures.

UNIT-II

[8 Hrs]

Operations Management: Plant Location Principles and types of plant Layout, Work study, Materials Management: Objectives - Need for inventory control- Inventory control techniques EOQ, ABC, HML, SDE, VED and FSN analysis.

UNIT-III

[12Hrs]

Human Resources Management (HRM): Concepts of HRM, Basic functions of HR manager, Job Evaluation and Merit Rating, Performance Appraisal, Methods of Performance appraisal Concepts Compensation.

Marketing Management: Functions of marketing, Marketing Mix, Marketing strategies based on Product life cycle, Channels of distribution (Place), Promotional

Mix.

UNIT-IV

[10 Hrs]

Project Management (PERT/CPM): Network analysis, Program Evaluation and Review Technique (PERT), Critical path method (CPM) - Identifying critical path, Difference between PERT & CPM (simple problems).

UNIT-V

[8 Hrs]

Strategic Management: Mission, Goals, objectives, policy, strategy, Environmental scanning, SWOT analysis, Steps in strategy formulation and implementation Generic strategy alternatives.

Course Outcomes: Upon completing the course, student will be able to

COs	Description	Blooms Level
CO 1	Understand the fundamentals of Management with specific insight as its function and role	Understanding
CO 2	Learn the concepts of production, Management of human Resources and Management of Marketing activities along with business environment	Understanding
CO 3	Apply the problem solving skills to demonstrate logical solution to real life problems	Applying
CO 4	Create the awareness of business strategies to deal with the dynamic business environment	Creating

Mapping of COs to POs:

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
CO 1	-	-	-	-	-	-	-	-	2	-	-	-	-	-
CO 2	-	-	-	-	-	2	-	-	-	-	-	-	-	-
CO 3	-	-	-	-	-	-	-	-	-	-	2	-	-	-
CO 4	-	-	-	-	-	-	-	-	-	-	2	-	-	-

Text Books:

T1. Dr. Arya Sri, "Management Science",TMH 2011.

T2. L.M. Prasad, "Principles & Practices of Management" Sultan chand& Sons, 2007.

Reference Books:

R1. K. Aswathappa and K. Sridhara Bhat, "Production and Operations Management", Himalaya Publishing House, 2010.

R2. Philip Kotler [Philip Kotler](#), [Kevin Keller](#), [Mairead Brady](#), [Malcolm Goodman](#), [Torben Hansen](#), "Marketing Management" Pearson Education Limited, 2016.

AMATEUR RADIO LAB

Subject Code: UGEC7K1822

IV Year/ II Semester

L	T	P	C
1	0	2	2

Prerequisites

- Analog and Digital Communication
- Electromagnetic Waves

Laboratory objectives: The objectives of this laboratory are

- To introduce exposure to the students about the different Radio facilities and their operating frequency regimes.
- To introduce Morse Coding for Amateur Radio Communication.

Experiments (Any 10 Experiment)

1. Study of working Principles in Community Radio
2. Observations in Transmission and Reception of Radio Signals
3. Study of Frequency allocations permitted for HAM Communication
4. Study of Various Modulation techniques like A3E, A1H, J3E and F3E
5. Study and design of Dipole Antenna for HAM Communication (7.00 -7.20 MHz)
6. Transmission of Message using Morse Code key Transmitter (8WPM)
7. Reception of Message using Morse Code key (8WPM)
8. Study and Observations of S-Band Ground Station Operations
9. Tracking of satellites using S-Band Ground Station
10. Analyzing and decoding the satellite data
11. Study on decoding and decryption techniques on Satellite Data

LOS	Description	Bloom's Level
LO 1	Study on working principles of Community and Amateur Radios	II – Understanding
LO 2	Illustrate various modulation techniques and frequency bands employed in HAM Communication	III-Applying
LO 3	Tracking and analysis of satellite data through S-band Ground Station	VI-Creating

Mapping of Lo's to PO's

POs	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2
LO 1	3	3							3	3				
LO 2	3	3	3						3	3			3	
LO 3	3	3	3						3	3			2	